Homework Solutions: Lectures 9–11
Fourier Series, DTFS Properties, and Frequency Response

CEC 315 – Signals and Systems

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Problem 1: Eigenfunction Property and CT Fourier Series Coefficients

Part (a): Eigenfunction Property

Given: $h(t) = e^{-4t}u(t)$, input $x(t) = \cos(3t)$.

Step 1: Compute $H(j3)$ from the definition.
Since $h(t)$ is causal, the lower limit is 0:

$$H(j3) = \int_0^\infty e^{-4\tau}\, e^{-j3\tau}\, d\tau = \int_0^\infty e^{-(4+j3)\tau}\, d\tau$$

Evaluating the integral:

$$H(j3) = \left[\frac{e^{-(4+j3)\tau}}{-(4+j3)}\right]_0^\infty = 0 - \frac{1}{-(4+j3)} = \frac{1}{4+j3}$$

Step 2: Convert to rectangular form.
Multiply numerator and denominator by the complex conjugate $(4 - j3)$:

$$H(j3) = \frac{1}{4+j3} \cdot \frac{4-j3}{4-j3} = \frac{4-j3}{4^2+3^2} = \frac{4-j3}{25} = \frac{4}{25} - j\frac{3}{25}$$

Step 3: Convert to polar form.

$$|H(j3)| = \frac{1}{|4+j3|} = \frac{1}{\sqrt{4^2+3^2}} = \frac{1}{\sqrt{25}} = \boxed{\frac{1}{5} = 0.2}$$

$$\angle H(j3) = -\arctan\left(\frac{3}{4}\right) = \boxed{-36.87^\circ}$$

Step 4: Write the output.

By the eigenfunction property, $\cos(3t)$ enters and exits as the same cosine, only scaled and phase-shifted:

$$\boxed{y(t) = 0.2\cos(3t - 36.87^\circ)}$$

Key Point

The frequency did *not* change—only the amplitude (scaled from 1 to 0.2) and phase (shifted by -36.87). This is the hallmark of the eigenfunction property.

Part (b): Fourier Series Coefficients

Given: $x(t)$ periodic with $T = 4$, $\quad x(t) = 3$ for $0 \le t < 1$, $\quad x(t) = 0$ for $1 \le t < 4$.

(i) Fundamental frequency

$$\boxed{\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{4} = \frac{\pi}{2} \text{ rad/s}}$$

(ii) DC coefficient a_0

Step 1: Apply the average-value formula:

$$a_0 = \frac{1}{T}\int_T x(t)\,dt = \frac{1}{4}\int_0^1 3\,dt = \frac{1}{4}\cdot 3\cdot(1-0) = \boxed{\frac{3}{4} = 0.75}$$

Physical meaning: a_0 is the **average (DC) value** of $x(t)$ over one period. The signal equals 3 for 1/4 of the period and 0 for the rest, so its average is 3/4.

(iii) General expression for a_k ($k \neq 0$)

Step 1: Set up the analysis integral, integrating only where $x(t) \neq 0$:

$$a_k = \frac{1}{4}\int_0^1 3\,e^{-jk(\pi/2)t}\,dt = \frac{3}{4}\int_0^1 e^{-jk(\pi/2)t}\,dt$$

Step 2: Evaluate the integral:

$$a_k = \frac{3}{4}\left[\frac{e^{-jk(\pi/2)t}}{-jk(\pi/2)}\right]_0^1 = \frac{3}{4}\cdot\frac{e^{-jk\pi/2}-1}{-jk\pi/2} = \frac{3}{4}\cdot\frac{1-e^{-jk\pi/2}}{jk\pi/2}$$

Step 3: Simplify by multiplying through:

$$\boxed{a_k = \frac{3}{2jk\pi}\left(1 - e^{-jk\pi/2}\right), \quad k \neq 0}$$

(iv) Numerical values of a_1 and a_2

For $k = 1$:

Compute $e^{-j\pi/2} = \cos(\pi/2) - j\sin(\pi/2) = 0 - j = -j$. Therefore:

$$a_1 = \frac{3}{2j\pi}(1 - (-j)) = \frac{3(1+j)}{2j\pi}$$

Multiply numerator and denominator by $-j$:

$$a_1 = \frac{3(1+j)(-j)}{2j\pi(-j)} = \frac{3(-j+1)}{2\pi} = \frac{3(1-j)}{2\pi}$$

Polar form:

$$|a_1| = \frac{3\sqrt{1^2+1^2}}{2\pi} = \frac{3\sqrt{2}}{2\pi} \approx 0.675, \qquad \angle a_1 = \arctan\left(\frac{-1}{1}\right) = \boxed{-45^\circ}$$

$$\boxed{a_1 = \frac{3(1-j)}{2\pi} \approx 0.477 - 0.477j, \quad |a_1| = \frac{3\sqrt{2}}{2\pi} \approx 0.675, \quad \angle a_1 = -45^\circ}$$

For $k = 2$:

Compute $e^{-j2\cdot\pi/2} = e^{-j\pi} = -1$. Therefore:

$$a_2 = \frac{3}{4j\pi}(1 - (-1)) = \frac{3\cdot 2}{4j\pi} = \frac{3}{2j\pi}$$

Multiply by $-j/(-j)$:

$$a_2 = \frac{-3j}{2\pi}$$

Polar form:

$$\boxed{a_2 = \frac{-3j}{2\pi} \approx -0.477j, \quad |a_2| = \frac{3}{2\pi} \approx 0.477, \quad \angle a_2 = -90^\circ}$$

Part (c): Conjugate Symmetry Verification

Claim: For a real signal, $a_{-k} = a_k^*$.

Step 1: Compute a_{-1} directly from the general formula with $k = -1$.

$e^{-j(-1)\pi/2} = e^{j\pi/2} = j$. Therefore:

$$a_{-1} = \frac{3}{2(-j)\pi}(1-j) = \frac{3(1-j)}{-2j\pi}$$

Multiply by $-j/(-j)$:

$$a_{-1} = \frac{3(1-j)(-j)}{2\pi} = \frac{3(-j+j^2)}{2\pi} = \frac{3(-j-1)}{2\pi} \quad \text{— wait, recalculate cleanly:}$$

$$(1-j)(-j) = -j + j^2 = -j + (-1) = -(1+j) \quad \Longrightarrow \quad a_{-1} = \frac{-3(1+j)}{-2\pi} = \frac{3(1+j)}{2\pi}$$

Step 2: Compute a_1^*:

$$a_1^* = \left(\frac{3(1-j)}{2\pi}\right)^* = \frac{3(1+j)}{2\pi}$$

Step 3: Compare:

$$a_{-1} = \frac{3(1+j)}{2\pi} = a_1^* \qquad \checkmark$$

Key Point

The conjugate symmetry property $a_{-k} = a_k^*$ holds for **any real-valued signal**. It is not a coincidence—it follows directly from taking the complex conjugate of the analysis integral when $x(t)$ is real.

Problem 2: CT Fourier Series — Convergence and Gibbs Phenomenon

Given: Odd square wave with $T = 2\pi$, $x(t) = +1$ for $0 < t < \pi$ and $x(t) = -1$ for $-\pi < t < 0$.

Part (a): Dirichlet Conditions

The three Dirichlet conditions for convergence of the CT Fourier series are:

1. **Absolute integrability over one period:** $\int_T |x(t)|\, dt < \infty$.
2. **Bounded variation:** $x(t)$ has a finite number of maxima and minima in any single period.
3. **Finite discontinuities:** $x(t)$ has a finite number of discontinuities in any single period, each with a finite jump.

Verification for the square wave:

1. $\int_{-\pi}^{\pi} |x(t)|\, dt = \int_{-\pi}^{\pi} 1\, dt = 2\pi < \infty$. ✓
2. In one period $(-\pi, \pi)$, the signal has no maxima or minima (it is piecewise constant), so the number of extrema is zero—certainly finite. ✓
3. There are exactly two discontinuities per period: at $t = 0$ and at $t = \pm\pi$. Each jump has size $|1 - (-1)| = 2$, which is finite. ✓

All three Dirichlet conditions are satisfied, so the Fourier series converges.

At each discontinuity, the Fourier series converges to the **average of the left-hand and right-hand limits**. For example, at $t = 0$:

$$\text{Fourier series at } t = 0 = \frac{x(0^-) + x(0^+)}{2} = \frac{-1+1}{2} = 0$$

Part (b): Convergence at Specific Points

At $t = 0$ (discontinuity):

By the Dirichlet theorem, the series converges to the midpoint of the jump:

$$\boxed{x_{\text{FS}}(0) = \frac{-1+1}{2} = 0}$$

Note: the actual signal is undefined at $t = 0$ (or defined as 0 by convention), but the Fourier series converges to 0 regardless.

At $t = \pi/2$ (continuity point):

Since $x(t)$ is continuous at $t = \pi/2$ (with value $+1$), the series converges to $x(\pi/2) = 1$.
Numerical verification using the first three nonzero terms:

$$x(t) \approx \frac{4}{\pi}\left[\sin t + \frac{\sin 3t}{3} + \frac{\sin 5t}{5}\right]$$

Substituting $t = \pi/2$ (so $\sin(\pi/2) = 1$, $\sin(3\pi/2) = -1$, $\sin(5\pi/2) = 1$):

$$x_3(\pi/2) = \frac{4}{\pi}\left[1 + \frac{-1}{3} + \frac{1}{5}\right] = \frac{4}{\pi}\left[\frac{15-5+3}{15}\right] = \frac{4}{\pi}\cdot\frac{13}{15} = \frac{52}{15\pi} \approx \frac{52}{47.12} \approx 1.103$$

The three-term approximation gives ≈ 1.103, which overshoots the exact value of 1 by about 10.3%. With more terms, this converges to 1.

Part (c): Partial Sums and Gibbs Phenomenon

(i) Explicit partial sums

$x_1(t)$ (1 odd harmonic, up to $k = 1$, i.e., $2N - 1 = 1$):

$$\boxed{x_1(t) = \frac{4}{\pi}\sin(t)}$$

$x_3(t)$ (3 odd harmonics, up to $k = 5$, i.e., $2N - 1 = 5$):

$$\boxed{x_3(t) = \frac{4}{\pi}\left[\sin(t) + \frac{\sin(3t)}{3} + \frac{\sin(5t)}{5}\right]}$$

(ii) Gibbs overshoot

The Gibbs phenomenon states that near each jump discontinuity, the partial sum overshoots by approximately **8.9**% of the jump height (roughly 9%), regardless of how many harmonics are included.

No, the overshoot does *not* disappear as $N \to \infty$. The overshooting region becomes narrower and narrows toward the discontinuity, but its peak amplitude remains at $\approx 9\%$ above the true value no matter how many terms are summed. Only in the limit of infinitely many terms does the energy in the error go to zero—but the pointwise overshoot persists at any finite N.

(iii) Why a finite sum cannot reconstruct a discontinuity

Each harmonic $\sin(kt)$ is a smooth, continuous, infinitely differentiable function. A finite linear combination of smooth functions is itself smooth and continuous—it cannot produce the sharp, instantaneous jump of a discontinuous signal. Only an infinite sum of harmonics, each contributing a small correction, can approximate an actual discontinuity.

Part (d): Smoothness and Spectral Decay

Signal	Spectral decay	Smoothness
Square wave	$\lvert a_k \rvert \sim 1/k$	Discontinuous (jump at $t = 0, \pi$)
Triangular wave	$\lvert a_k \rvert \sim 1/k^2$	Continuous, but dx/dt is discontinuous

The **triangular wave is smoother**. The general rule is:

Key Point

The smoother a signal is, the faster its Fourier coefficients decay. Specifically:

- A signal with a **jump discontinuity** (0 continuous derivatives) has coefficients decaying as $1/k$.
- A signal that is **continuous** but has a discontinuous first derivative has coefficients decaying as $1/k^2$.
- More generally, if the first discontinuous derivative is the mth, the coefficients decay as $1/k^{m+1}$.

Fewer high-frequency harmonics are needed to represent a smoother signal.

Problem 3: CT Fourier Series Properties

Given: $x(t)$ real-valued, $T = \pi$, $\omega_0 = 2$ rad/s, with coefficients:

$$a_0 = 2, \quad a_{\pm 1} = 1, \quad a_{\pm 2} = \tfrac{1}{4}, \quad a_{\pm 3} = \tfrac{1}{9}, \quad a_k = 0 \text{ for } |k| > 3$$

Part (a): Linearity

(i) Fourier coefficients of $\sin(2t)$

Using Euler's relation with $\omega_0 = 2$:

$$\sin(2t) = \frac{e^{j2t} - e^{-j2t}}{2j} = \frac{1}{2j} e^{j\omega_0 t} - \frac{1}{2j} e^{-j\omega_0 t}$$

Comparing with the synthesis equation $\sum_k b_k \, e^{jk\omega_0 t}$:

$$\boxed{b_1 = \frac{1}{2j} = -\frac{j}{2}, \qquad b_{-1} = -\frac{1}{2j} = \frac{j}{2}, \qquad b_k = 0 \text{ for all other } k}$$

(ii) Coefficients of $w(t) = 4x(t) + 6\sin(2t)$

By the linearity property, $c_k = 4a_k + 6b_k$:

k	$4a_k$	$6b_k$	$c_k = 4a_k + 6b_k$
0	$4(2) = 8$	0	$\boxed{8}$
1	$4(1) = 4$	$6(-j/2) = -3j$	$\boxed{4 - 3j}$
-1	$4(1) = 4$	$6(j/2) = 3j$	$\boxed{4 + 3j}$
2	$4(1/4) = 1$	0	$\boxed{1}$
-2	$4(1/4) = 1$	0	$\boxed{1}$
3	$4(1/9) = 4/9$	0	$\boxed{4/9}$
-3	$4(1/9) = 4/9$	0	$\boxed{4/9}$
$\lvert k\rvert > 3$	0	0	$\boxed{0}$

Key Point

Note that $c_{-1} = (c_1)^* = (4 - 3j)^* = 4 + 3j$ as expected for a real signal ($w(t)$ is real).

Part (b): Time-Shifting Property

Given: $y(t) = x(t - \pi/4)$, so $t_0 = \pi/4$, $\omega_0 = 2$.

(i) New coefficients b_k

By the time-shifting property:

$$b_k = a_k\, e^{-jk\omega_0 t_0} = a_k\, e^{-jk\cdot 2\cdot(\pi/4)} = \boxed{a_k\, e^{-jk\pi/2}}$$

Explicitly:

k	a_k	$e^{-jk\pi/2}$	$b_k = a_k\, e^{-jk\pi/2}$
0	2	1	2
± 1	1	$e^{\mp j\pi/2} = \mp j$	$\mp j$
± 2	1/4	$e^{\mp j\pi} = -1$	$-1/4$
± 3	1/9	$e^{\mp j3\pi/2} = \pm j$	$\pm j/9$

(ii) Magnitude verification

For $k = 1$:

$$|b_1| = |a_1| \cdot |e^{-j\pi/2}| = 1 \cdot 1 = 1 = |a_1| \qquad \checkmark$$

For $k = 2$:

$$|b_2| = |a_2| \cdot |e^{-j\pi}| = \tfrac{1}{4} \cdot 1 = \tfrac{1}{4} = |a_2| \qquad \checkmark$$

(iii) Implication for power spectral density

Since $|b_k| = |a_k|$ for all k, the power spectrum $|b_k|^2 = |a_k|^2$ is identical. By Parseval's theorem, $y(t)$ has the **same average power** as $x(t)$. Time shifting only changes phases—it never redistributes power across frequencies.

Part (c): Differentiation Property

Given: $g(t) = dx(t)/dt$ with $d_k = jk\omega_0\, a_k = jk(2)\, a_k = 2jk\, a_k$.

k	a_k	$d_k = 2jk\, a_k$	$\lvert d_k\rvert$
0	2	0	0
± 1	1	$\pm 2j$	2
± 2	1/4	$\pm j$	1
± 3	1/9	$\pm \frac{2j}{3}$	2/3

$$\boxed{|d_1| = 2, \qquad |d_2| = 1, \qquad |d_3| = \frac{2}{3}}$$

Observation: The ratio $|d_k|/|a_k| = 2|k|$ grows linearly with k. Differentiation multiplies each harmonic's amplitude by $k\omega_0$, so **higher harmonics are amplified more than lower**

ones. A signal that was dominated by low frequencies will have its derivative dominated more by higher frequencies. This is why differentiation "sharpens" a signal and amplifies noise.

Part (d): Parseval's Theorem

Step 1: Write out all nonzero $|a_k|^2$ terms:

$$P_x = \sum_{k=-\infty}^{+\infty} |a_k|^2 = |a_0|^2 + 2\big(|a_1|^2 + |a_2|^2 + |a_3|^2\big)$$

Step 2: Substitute values:

$$P_x = (2)^2 + 2\left[(1)^2 + \left(\frac{1}{4}\right)^2 + \left(\frac{1}{9}\right)^2\right] = 4 + 2\left[1 + \frac{1}{16} + \frac{1}{81}\right]$$

Step 3: Evaluate numerically:

$$P_x = 4 + 2\,[1.000 + 0.0625 + 0.01235] = 4 + 2(1.07485) = 4 + 2.1497 = \boxed{6.1497}$$

Exact form:

$$P_x = 4 + 2 + \frac{1}{8} + \frac{2}{81} = 6 + \frac{1}{8} + \frac{2}{81} = 6 + \frac{81+16}{648} = 6\frac{97}{648}$$

DC fraction:

$$\frac{|a_0|^2}{P_x} = \frac{4}{6.1497} \approx \boxed{65.0\%}$$

The DC component alone carries about 65% of the total signal power.

Problem 4: Discrete-Time Fourier Series

Given: $x[n]$ periodic with $N = 6$; $x[n] = 1$ for $n = 0, 1, 2$ and $x[n] = 0$ for $n = 3, 4, 5$.

Part (a): DTFS Fundamentals

The **DTFS pair** for a signal with period N is:

Analysis equation:

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n]\, e^{-jk(2\pi/N)n}$$

Synthesis equation:

$$x[n] = \sum_{k=\langle N \rangle} a_k\, e^{jk(2\pi/N)n}$$

A DT periodic signal with period N has exactly **N distinct** Fourier coefficients. This is because the DT complex exponentials $e^{jk(2\pi/N)n}$ and $e^{j(k+N)(2\pi/N)n}$ are **identical** for integer n (since $e^{j2\pi n} = 1$). Therefore, adding more harmonics beyond the Nth does not produce new signals—they simply repeat. Unlike the CT case, there are no convergence issues because the DTFS is a finite sum.

Part (b): Computing a_0, a_1, a_2, a_3

With $N = 6$, the fundamental frequency is $\omega_0 = 2\pi/6 = \pi/3$. The analysis sum runs from $n = 0$ to 5, but only $n = 0, 1, 2$ contribute (where $x[n] = 1$):

$$a_k = \frac{1}{6} \sum_{n=0}^{2} e^{-jk(\pi/3)n} = \frac{1}{6}\left(1 + e^{-jk\pi/3} + e^{-j2k\pi/3}\right)$$

For $k = 0$:

$$a_0 = \frac{1}{6}(1 + 1 + 1) = \boxed{\frac{1}{2}}$$

For $k = 1$:

Let $r = e^{-j\pi/3} = \cos(\pi/3) - j\sin(\pi/3) = \dfrac{1}{2} - j\dfrac{\sqrt{3}}{2}$.

Compute each term:

$$e^{-j\pi/3} = \tfrac{1}{2} - j\tfrac{\sqrt{3}}{2}$$

$$e^{-j2\pi/3} = \cos(2\pi/3) - j\sin(2\pi/3) = -\tfrac{1}{2} - j\tfrac{\sqrt{3}}{2}$$

Sum: $1 + \left(\frac{1}{2} - j\frac{\sqrt{3}}{2}\right) + \left(-\frac{1}{2} - j\frac{\sqrt{3}}{2}\right) = 1 - j\sqrt{3}$

$$\boxed{a_1 = \frac{1 - j\sqrt{3}}{6}, \qquad |a_1| = \frac{\sqrt{1+3}}{6} = \frac{2}{6} = \frac{1}{3}, \qquad \angle a_1 = -\arctan(\sqrt{3}) = -60^\circ}$$

For $k = 2$:

$$e^{-j2\pi/3} = -\tfrac{1}{2} - j\tfrac{\sqrt{3}}{2}$$
$$e^{-j4\pi/3} = \cos(4\pi/3) - j\sin(4\pi/3) = -\tfrac{1}{2} + j\tfrac{\sqrt{3}}{2}$$

Sum: $1 + \left(-\frac{1}{2} - j\frac{\sqrt{3}}{2}\right) + \left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right) = 1 - 1 = 0$

$$\boxed{a_2 = 0}$$

For $k = 3$:

$$e^{-j3\pi/3} = e^{-j\pi} = -1$$
$$e^{-j6\pi/3} = e^{-j2\pi} = 1$$

Sum: $1 + (-1) + 1 = 1$

$$\boxed{a_3 = \frac{1}{6}, \qquad |a_3| = \frac{1}{6}, \qquad \angle a_3 = 0^\circ}$$

Part (c): Periodicity of Coefficients

Since $a_{k+N} = a_k$ with $N = 6$:

$$a_4 = a_{4-6} = a_{-2} = a_2^* = 0^* = \boxed{0}$$

$$a_5 = a_{5-6} = a_{-1} = a_1^* = \frac{1 + j\sqrt{3}}{6} \approx \boxed{\frac{1 + j\sqrt{3}}{6}}, \qquad |a_5| = \frac{1}{3}$$

Why are DTFS coefficients periodic? Because multiplying the synthesis exponential $e^{jk(2\pi/N)n}$ by $e^{j2\pi n} = 1$ (which is always 1 for integer n) shows that $e^{j(k+N)(2\pi/N)n} = e^{jk(2\pi/N)n}$. The $(k+N)$th harmonic is *exactly the same signal* as the kth harmonic. There is therefore no new information in a_{k+N} beyond a_k.

Part (d): Parseval's Theorem Verification

Left-hand side (time domain):

$$\frac{1}{N}\sum_{n=0}^{5} |x[n]|^2 = \frac{1}{6}(1^2 + 1^2 + 1^2 + 0^2 + 0^2 + 0^2) = \frac{3}{6} = \boxed{\frac{1}{2}}$$

Right-hand side (frequency domain):

$$\sum_{k=0}^{5} |a_k|^2 = |a_0|^2 + |a_1|^2 + |a_2|^2 + |a_3|^2 + |a_4|^2 + |a_5|^2$$

$$= \left(\frac{1}{2}\right)^2 + \left(\frac{1}{3}\right)^2 + 0^2 + \left(\frac{1}{6}\right)^2 + 0^2 + \left(\frac{1}{3}\right)^2$$

$$= \frac{1}{4} + \frac{1}{9} + 0 + \frac{1}{36} + 0 + \frac{1}{9}$$

Finding a common denominator of 36:

$$= \frac{9}{36} + \frac{4}{36} + \frac{1}{36} + \frac{4}{36} = \frac{18}{36} = \boxed{\frac{1}{2}} \qquad \checkmark$$

Both sides equal $\frac{1}{2}$. Parseval's theorem is verified.

Problem 5: Frequency Response and Filtering

Given: $h(t) = e^{-3t}u(t)$, input $x(t) = 2 + 4\cos(3t) + 2\cos(9t)$.

Part (a): Frequency Response $H(j\omega)$

Step 1: Apply the definition with causal $h(t)$:

$$H(j\omega) = \int_0^\infty e^{-3\tau}\, e^{-j\omega\tau}\, d\tau = \int_0^\infty e^{-(3+j\omega)\tau}\, d\tau$$

Step 2: Evaluate (the integral converges since $\mathrm{Re}(3 + j\omega) = 3 > 0$):

$$H(j\omega) = \left[\frac{e^{-(3+j\omega)\tau}}{-(3+j\omega)}\right]_0^\infty = 0 + \frac{1}{3+j\omega}$$

$$\boxed{H(j\omega) = \frac{1}{3+j\omega}, \qquad a = 3}$$

Part (b): Evaluate at Each Harmonic

The frequencies present in $x(t)$ are $\omega = 0$, $\omega = 3$, and $\omega = 9$ rad/s.

At $\omega = 0$:

$$H(0) = \frac{1}{3+0} = \frac{1}{3}, \qquad |H(0)| = \frac{1}{3}, \qquad \angle H(0) = 0^\circ$$

At $\omega = 3$:

$$H(j3) = \frac{1}{3+j3} = \frac{1}{3(1+j)}$$

$$|H(j3)| = \frac{1}{3|1+j|} = \frac{1}{3\sqrt{2}} = \frac{\sqrt{2}}{6} \approx 0.2357$$

$$\angle H(j3) = -\arctan\left(\frac{3}{3}\right) = -\arctan(1) = -45^\circ$$

At $\omega = 9$:

$$H(j9) = \frac{1}{3+j9} = \frac{1}{3(1+j3)}$$

$$|H(j9)| = \frac{1}{3|1+j3|} = \frac{1}{3\sqrt{1^2+3^2}} = \frac{1}{3\sqrt{10}} \approx 0.1054$$

$$\angle H(j9) = -\arctan\left(\frac{9}{3}\right) = -\arctan(3) \approx -71.57^\circ$$

Completed table:

ω (rad/s)	$H(j\omega)$	$\lvert H(j\omega)\rvert$	$\angle H(j\omega)$
0	$\frac{1}{3}$	$\frac{1}{3} \approx 0.333$	$0°$
3	$\frac{1}{3(1+j)}$	$\frac{1}{3\sqrt{2}} \approx 0.236$	$-45°$
9	$\frac{1}{3(1+j3)}$	$\frac{1}{3\sqrt{10}} \approx 0.105$	$-71.57°$

Part (c): Output Signal $y(t)$

Apply the eigenfunction property to each component of $x(t)$:

DC term ($\omega = 0$):

$$2 \xrightarrow{\times H(0)} 2 \cdot \frac{1}{3} = \frac{2}{3}$$

Fundamental ($\omega = 3$):

$$4\cos(3t) \to 4 \cdot \frac{1}{3\sqrt{2}} \cos(3t - 45°) = \frac{4}{3\sqrt{2}} \cos(3t - 45°) = \frac{2\sqrt{2}}{3} \cos(3t - 45°)$$

Third harmonic ($\omega = 9$):

$$2\cos(9t) \to 2 \cdot \frac{1}{3\sqrt{10}} \cos(9t - 71.57°) = \frac{2}{3\sqrt{10}} \cos(9t - 71.57°)$$

Complete output:

$$\boxed{y(t) = \frac{2}{3} + \frac{2\sqrt{2}}{3} \cos(3t - 45°) + \frac{2}{3\sqrt{10}} \cos(9t - 71.57°)}$$

$$\approx 0.667 + 0.943\cos(3t - 45°) + 0.211\cos(9t - 71.57°)$$

Part (d): Filtering Interpretation

(i) Amplitude ratio

At $\omega = 3$: the input amplitude is 4, the output amplitude is $4 \cdot \frac{1}{3\sqrt{2}} \approx 0.943$.

At $\omega = 9$: the input amplitude is 2, the output amplitude is $2 \cdot \frac{1}{3\sqrt{10}} \approx 0.211$.

Ratio of output amplitudes at the two frequencies:

$$\frac{|H(j9)|}{|H(j3)|} = \frac{1/(3\sqrt{10})}{1/(3\sqrt{2})} = \frac{\sqrt{2}}{\sqrt{10}} = \frac{1}{\sqrt{5}} \approx 0.447$$

The $\omega = 9$ component is attenuated by an additional factor of $1/\sqrt{5} \approx 0.447$ relative to the $\omega = 3$ component. **The higher harmonic is attenuated more.**

(ii) Lowpass or highpass?

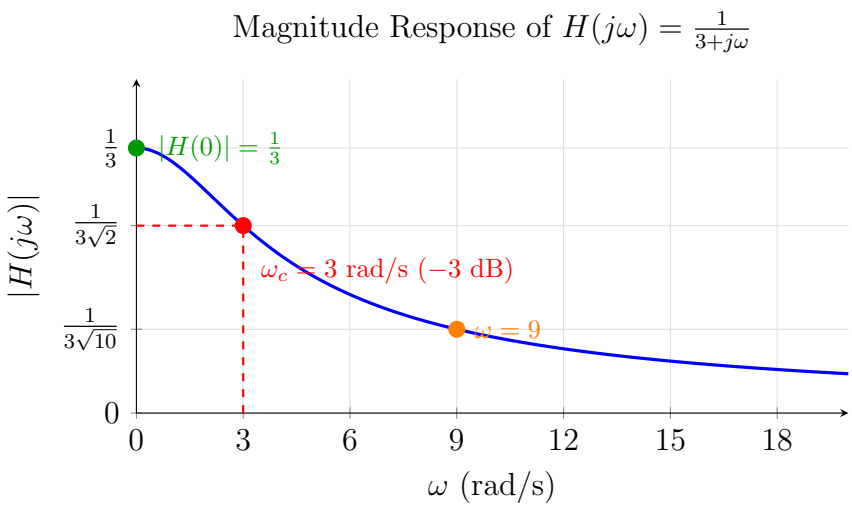
From the table, $|H(j\omega)|$ **decreases** as ω increases ($0.333 \to 0.236 \to 0.105$). This system passes low frequencies and attenuates high frequencies: it is a **lowpass filter**.

(iii) Sketch of $|H(j\omega)|$ and cutoff frequency

The -3 dB cutoff is found by setting $|H(j\omega_c)| = \dfrac{1}{\sqrt{2}}|H(0)|$:

$$\frac{1}{\sqrt{3^2+\omega_c^2}} = \frac{1}{\sqrt{2}} \cdot \frac{1}{3} \quad \Longrightarrow \quad \sqrt{3^2+\omega_c^2} = 3\sqrt{2} \quad \Longrightarrow \quad 9+\omega_c^2 = 18 \quad \Longrightarrow \quad \omega_c = 3 \text{ rad/s}$$

Note that $\omega = 3$ rad/s is *exactly* the cutoff frequency, which is why $\angle H(j3) = -45$ and $|H(j3)| = 1/(3\sqrt{2})$.



Part (e): Effect of the Pole

(i) Change in cutoff frequency

For $h'(t) = e^{-10t}u(t)$, the frequency response is $H'(j\omega) = \dfrac{1}{10+j\omega}$.

The cutoff frequency is $\omega_c' = 10$ rad/s, compared to $\omega_c = 3$ rad/s for the original system. **The cutoff frequency increases from 3 to 10 rad/s.**

(ii) Similarity of output to input

A higher cutoff frequency means a **wider bandwidth**: the system passes a broader range of frequencies with less attenuation. At $\omega = 3$ and $\omega = 9$ rad/s, both harmonics are now well within the passband, so both are passed with relatively little attenuation. The output $y'(t)$ would be **more similar to the input** $x(t)$—the relative amplitudes of the harmonics change much less than they did with the original $a = 3$ system.

Key Point

General rule: increasing the parameter a in $h(t) = e^{-at}u(t)$ moves the pole further into the left half-plane, widening the bandwidth. The system acts as a "less aggressive" lowpass filter, allowing more of the input's spectral content to pass through.

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